

Abasyn University

Department of Computing

Assignment #2

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Question 1

What is meant by Hybrid plants?

When we look at hybrid plants, we're basically talking about plants that are the result of cross-pollinating two different parent plants. These parent plants usually come from distinct varieties, species, or even genera. The whole point of creating a hybrid is to combine desirable traits from both parents into a single offspring. This is a pretty common practice in agriculture and horticulture, and it's something scientists have been doing for ages to get better crops.

What's interesting is that these hybrids often show what's called "hybrid vigor," or heterosis. This means they can sometimes be stronger, grow faster, be more resistant to diseases, or produce more fruit or flowers than either of their parent plants. It's like they get the best of both worlds and then some. This happens because the genes from the two parents can interact in ways that boost the overall performance of the plant.

Think about it like this: imagine you have one parent plant that's really good at surviving dry spells but doesn't produce much fruit, and another parent that produces tons of fruit but wilts easily without constant watering. If you cross them, you might get a hybrid plant that can handle some drought and still give you a decent harvest. That's the kind of thing breeders are aiming for. It's not just random; it's a deliberate process to improve on what nature provided.

This process can be done naturally through pollination, or it can be done artificially by humans in a lab or field. For instance, many of the corn varieties we eat today are hybrids, bred to have higher yields and better disease resistance. It's a really effective way to develop new plant varieties that are better suited for specific environments or for human needs.